

**Lab Report No _5_
Led Pattern generation using EdSim51 Simulator**

Embedded Systems



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Lab Task No 1:

Solution:

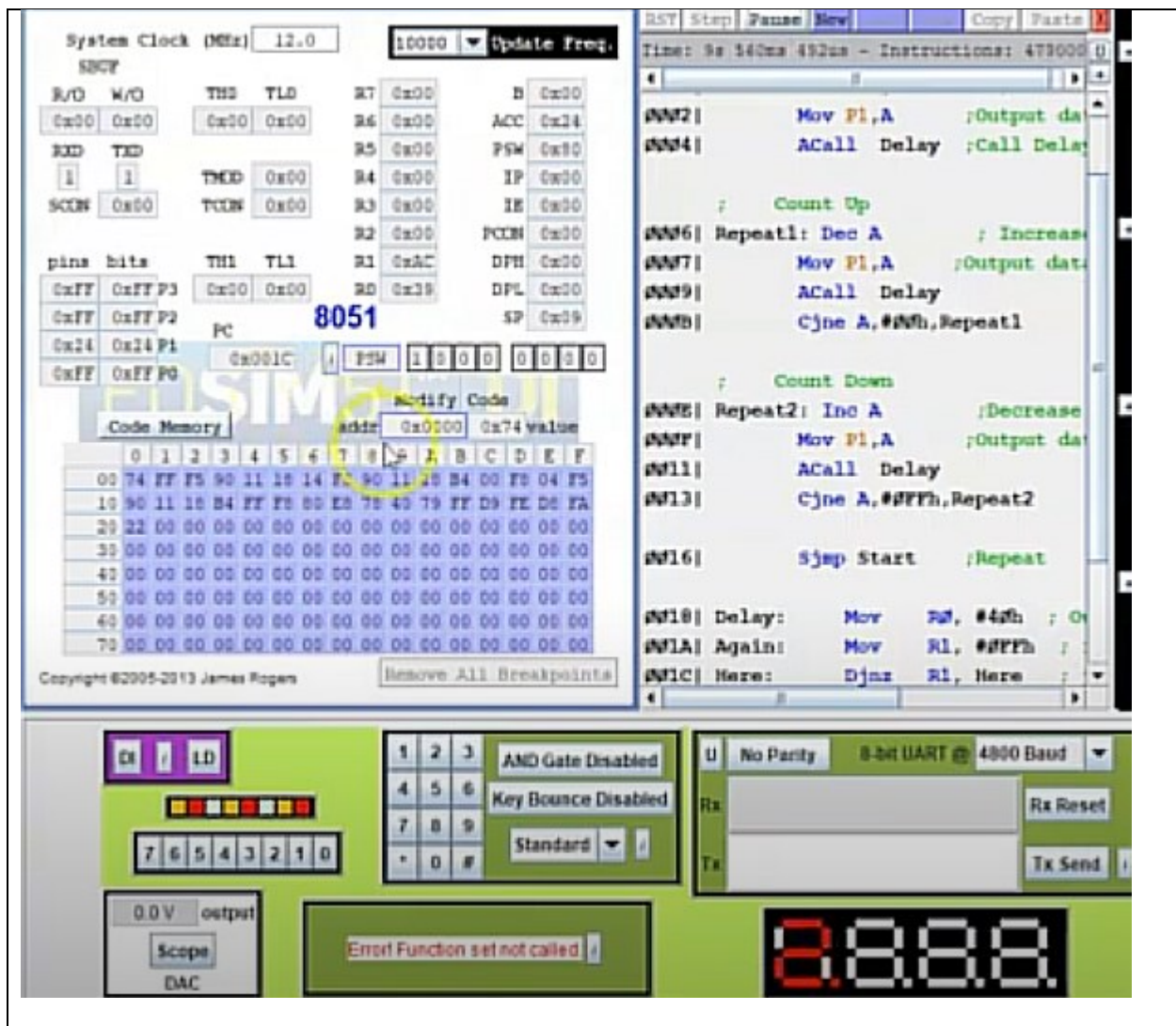
Brief description (3-5 lines)
Write a program which act as up and down counter from 0 to 9 and 9 to 0 using the 8- LEDs.
Code
<pre>ORG 0000H LJMP MAIN ORG 0030H MAIN: MOV P1, #0FFH ; Initialize Port 1 as output MOV R0, #0 ; Counter value MOV R1, #0 ; Direction flag (0=up, 1=down) LOOP: MOV A, R0 ; Move counter value to accumulator CALL DISPLAY ; Convert and display number CALL DELAY ; Add delay MOV A, R1 ; Check direction JZ COUNT_UP ; If R1=0, count up COUNT_DOWN: DJNZ R0, LOOP ; Decrement and continue if not zero MOV R1, #0 ; Change direction to up MOV R0, #0 ; Reset counter to 0 SJMP LOOP COUNT_UP: INC R0 ; Increment counter CJNE R0, #10, LOOP ; Continue if not 10 MOV R1, #1 ; Change direction to down MOV R0, #9 ; Set counter to 9 SJMP LOOP DISPLAY: MOV DPTR, #PATTERN ; Load pattern table address MOVC A, @A+DPTR ; Get pattern for current number</pre>

```

MOV P1, A      ; Output to LEDs
RET
DELAY:
MOV R6, #5     ; Outer loop counter
DELAY1:
MOV R7, #200   ; Inner loop counter
DELAY2:
DJNZ R7, DELAY2 ; Decrement inner counter
DJNZ R6, DELAY1 ; Decrement outer counter
RET
PATTERN:      ; LED patterns for numbers 0-9
DB 11000000B ; 0
DB 11111001B ; 1
DB 10100100B ; 2
DB 10110000B ; 3
DB 10011001B ; 4
DB 10010010B ; 5
DB 10000010B ; 6
DB 11111000B ; 7
DB 10000000B ; 8
DB 10010000B ; 9
END

```

The results (Screenshot)



Lab Task No 2:

Solution:

Brief description (3-5 lines)

Write a program which turn on the rightmost led by pressing the leftmost switch and so on.

Code

```

ORG 0H
SJMP MAIN
ORG 30H
MAIN:

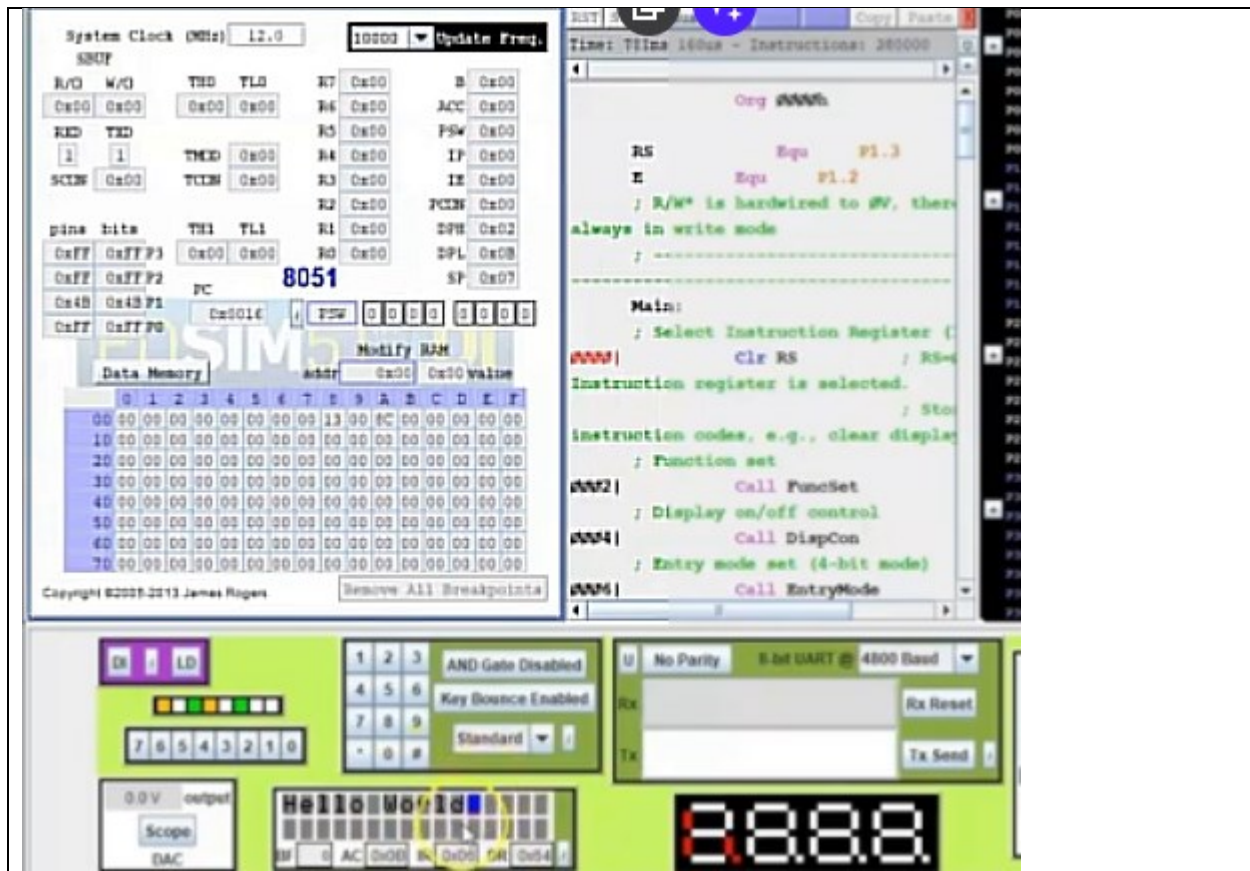
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```

; Initialize Port 1 for LEDs (Output)
MOV P1, #0FFH
; Main program loop
LOOP:
; Read switches from P0
MOV A, P0
; Reverse the bits using rotate operations
MOV B, #08H ; Counter for 8 bits
MOV R0, #00H ; Initialize result register
REVERSE:
RRC A ; Rotate right through carry
MOV C, ACC.0 ; Get the bit that was rotated
MOV A, R0 ; Get the result so far
RLC A ; Rotate left through carry
MOV R0, A ; Store the result
MOV A, B ; Get the counter
DEC A ; Decrement counter
MOV B, A ; Store the counter
MOV A, P0 ; Get the original input again
JNZ REVERSE ; If not done, continue reversing
; Output reversed value to LEDs
MOV A, R0 ; Get the reversed result
CPL A ; Complement (LEDs are active-low)
MOV P1, A ; Output to LEDs
SJMP LOOP ; Repeat forever
END

```

The results (Screenshot)



Lab Task No 3:

Solution:

Brief description (3-5 lines)

Write a program code which shows the number of key pressed on LED bar.

Code

```
ORG 0H
LJMP MAIN
ORG 100H
MAIN:
    ; Initialize counter in R0
    MOV R0, #0    ; R0 will store the count of key presses
    ; Configure Port 1 for LEDs (output)
    MOV P1, #0FFH ; Initially all LEDs are OFF
LOOP:
```

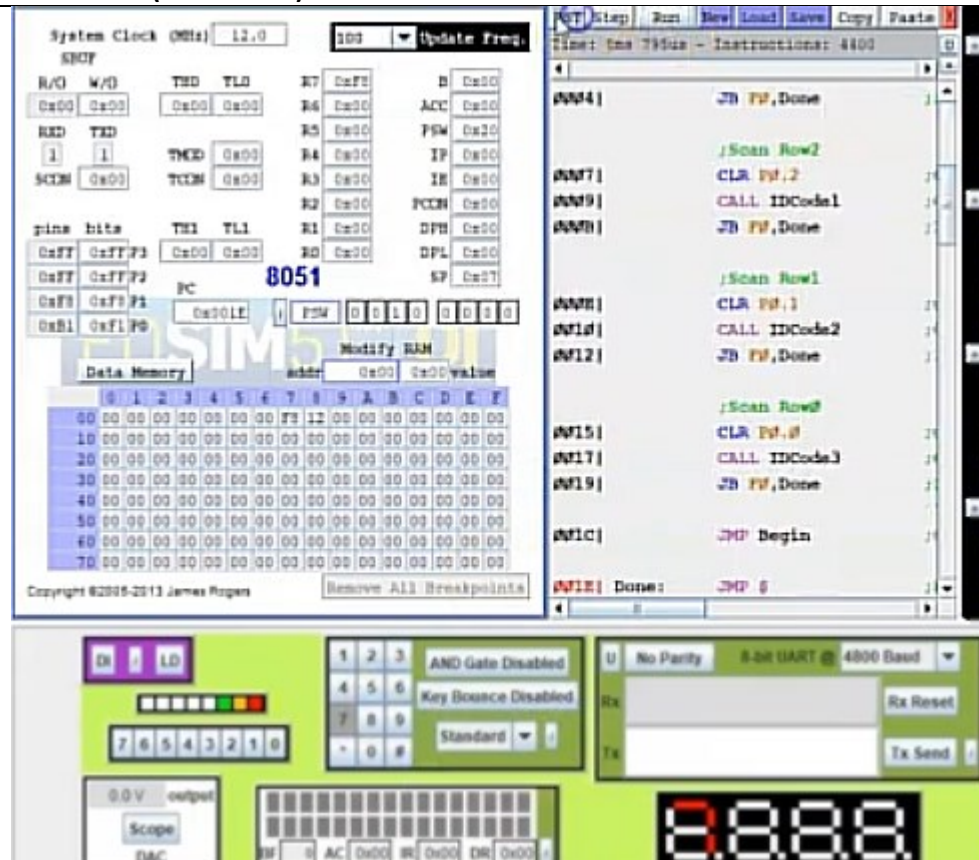
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; Check if key is pressed (P2.0)
JB P2.0, CHECK_RELEASE ; If key not pressed (P2.0 = 1),
check for release
SJMP LOOP ; If still pressed, keep checking
CHECK_RELEASE:
; Wait for key release
JNB P2.0, LOOP ; If key pressed (P2.0 = 0), go back to loop
; Key was released, increment counter
INC R0 ; Increment the counter
; Check if counter exceeded 8 (max LEDs)
MOV A, R0
CJNE A, #9, DISPLAY ; If not 9, continue to display
MOV R0, #0 ; If 9, reset counter to 0
DISPLAY:
; Convert counter to LED pattern
MOV A, R0 ; Move counter to accumulator
MOV B, #0FFH ; Start with all LEDs off
; Skip if counter is 0
JZ SHOW_LEDS
CONVERT_LOOP:
RR A ; Rotate right (will set carry if 1)
MOV B, A ; Move pattern to B
CPL B ; Complement B (0s become 1s and vice versa)
DJNZ ACC, CONVERT_LOOP ; Decrement A and continue if not
zero
SHOW_LEDS:
MOV P1, B ; Output pattern to LEDs
; Add delay for debouncing
ACALL DELAY
SJMP LOOP ; Return to main loop
DELAY:
MOV R7, #200
DELAY_OUTER:
MOV R6, #250
DELAY_INNER:

```


DJNZ R6, DELAY_INNER
DJNZ R7, DELAY_OUTER
RET
END

The results (Screenshot)



Conclusion

In conclusion, The EdSim51 Simulator was used to generate LED patterns, demonstrating the fundamental principles of 8051 microcontroller programming and I/O operations. The project demonstrated bit manipulation and delay routines, showcasing the simplicity and effectiveness of basic microcontroller concepts. This hands-on experience provides valuable insights into embedded systems programming and real-time pattern generation, laying a strong foundation for more complex microcontroller applications.